

Preventing White Spots

A Deep Dive Into Fluoride Releasing Adhesives

October 28, 2025

Introduction

Behavioral Obj.

Clinical Cases

Decision-Making

Thinking Hats

The Team



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What Do You Think?



Behavioral Objectives

Identify the biological and clinical factors that contribute to the formation of white spot lesions around orthodontic brackets.	1
Differentiate between conventional bonding agents and fluoride-releasing adhesives by explaining their composition, mechanism of action, and their clinical implications.	2
Describe additional preventive strategies used in orthodontics (e.g., fluoride varnishes, sealants, glass ionomer cements, patient hygiene protocols, bonding area) and analyze how they complement adhesive selection.	3
Compare and contrast the advantages and disadvantages of fluoride-releasing adhesives and adjunctive preventive methods using current scientific evidence.	4
Apply a clinical decision tree to select appropriate preventive measures for white spot lesion management based on patient risk factors and treatment context.	5

BO 1:

Identify the biological and clinical factors that contribute to the formation of white spot lesions around orthodontic brackets.

Behavioral Objective 1

Search Strategy

PICO

P- Patients undergoing fixed orthodontic treatment

I- Presence of orthodontic brackets

C- Patients without orthodontic appliances

O- Development of white spot lesions

Keywords

“White spot lesions” OR “enamel demineralization”

“Fixed orthodontic appliances” OR “orthodontic brackets”

“Fluoride exposure” OR “fluoride-releasing adhesives”

Database Used

PubMed

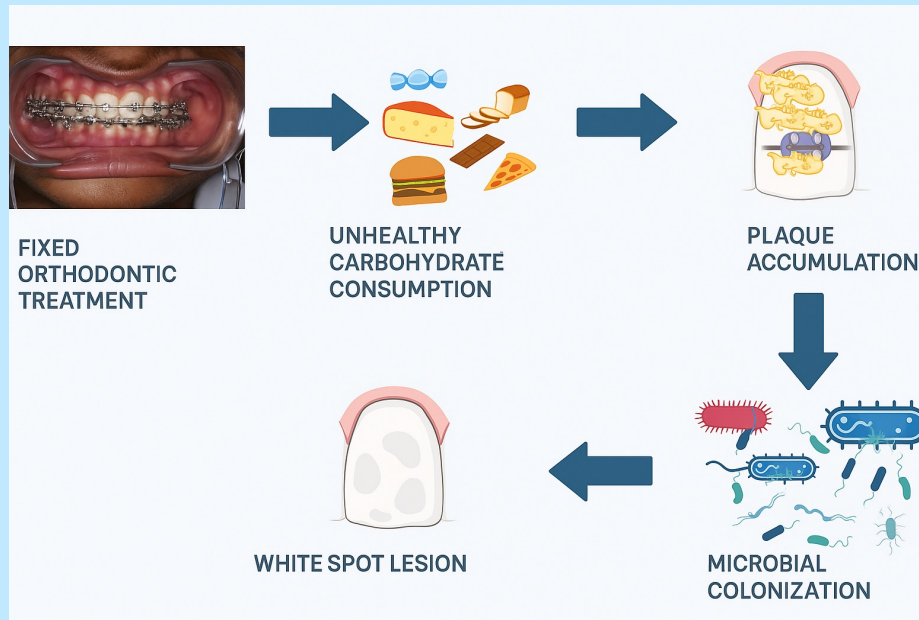
Highest Level of Evidence

Systematic Review

What causes white spots?

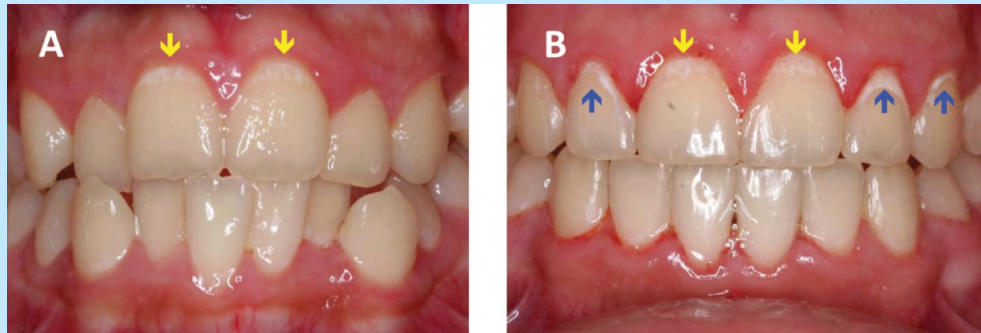
How it happens:

- ★ Plaque builds up around brackets
- ★ Bacteria ferments sugars, producing acids
- ★ pH drops below the critical level for enamel
- ★ Demineralization > remineralization causing white spot lesion



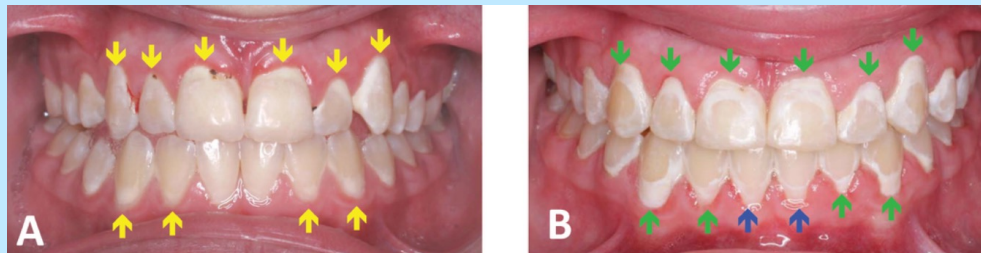
Prevalence:

- ★ Range 2-97% experience decalcification post-treatment with 15.5-40% WSL's present prior to treatment
- ★ 23% develop them during active treatment
- ★ Maxillary teeth are 2.5 times more affected than mandibular teeth



Yellow = existing white spots

Blue = new white spot lesions post-treatment



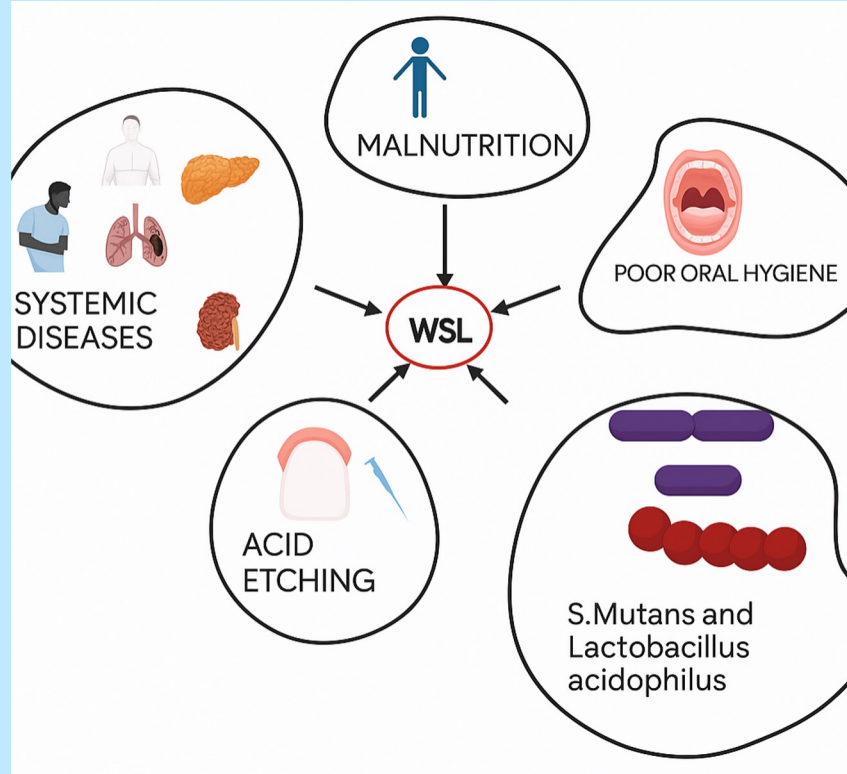
Yellow = existing white spots

Blue = new white spot lesions post-treatment

Green = worsened white spot lesions post-treatment

Biologic Factors

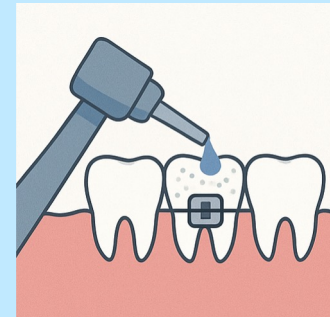
White Spot Lesions have many causes related to behavior, general health of the individual, and the oral microbiome.



Clinical Factors

- ★ Patient compliance:
 - Reinforce oral hygiene practices due to increased cleaning difficulty around brackets
 - Emphasize the importance of controlled dietary habits (limit fermentable carbohydrates).

- ★ Bracket and bonding procedures:
 - Increased etching times result in more demineralization, possibly increasing susceptibility to white spot formation
 - Bonding techniques and prevention methods (to come...)



Clinical Factors

- ★ Bonding material choice
 - Some materials may offer better fluoride release and protection against white spots.
- ★ Fluoride exposure
 - Maintain higher fluoride exposure through toothpaste, rinses, or varnish applications during orthodontic treatment.
- ★ Duration of treatment
 - Longer treatment time correlates with increased duration of plaque accumulation and white spot formation.



BO 2:

Differentiate between conventional bonding agents and fluoride-releasing adhesives by explaining their composition, mechanism of action and their clinical implications

Behavioral Objective 2

Search Strategy

PICO

P- Teeth requiring bonding for orthodontic procedures

I- Fluoride-releasing adhesives

C- Conventional bonding agents

O- Differences in clinical outcomes

Database Used

PubMed

Keywords

"Fluoride-releasing adhesives" OR "fluoride-containing bonding agents"

"Conventional bonding agents" OR "resin-based adhesives"

"Composition" OR "mechanism of action" OR

"polymerization"

"RMGI" AND "mechanism of action"

"Compomer" AND "mechanism of action"

"Giomer" AND "mechanism of action"

Highest Level of Evidence

Systematic Review

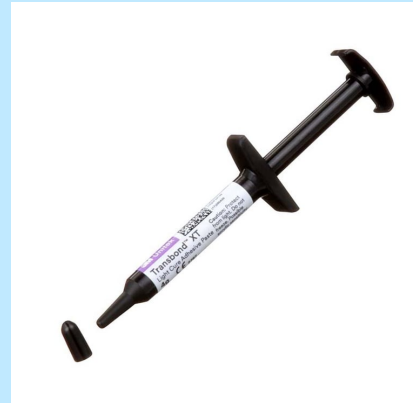
Conventional Resins used for Orthodontic Bonding

Example:

- Transbond XT

Composition:

- 37% orthophosphoric acid (ETCH-37TM, BISCO)
- Primer Bisphenol A Diglycidyl Ether Dimethacrylate(BISGMA), Triethylene Glycol Dimethacrylate (TEGDMA), 4-(Dimethylamino)-Benzeneethanol
- Paste: Silane Treated Quartz, BISGMA, Bisphenol A Dimethacrylate, Silane Treated Silica, Diphenyliodonium Hexafluorophosphate, Triphenylantimony



Tallani S (2023, November 7) Clinical performance of light-cured orthodontic adhesives for bonding brackets - an *in-vitro* study. F1000Res.;12:1442. doi: 10.12688/f1000research.141328.1. PMID: 38264264; PMCID: PMC10804051.
Condò R (2021 Jul 16).Physical and chemical mechanisms involved in adhesion of orthodontic bonding composites: in vitro evaluations. BMC Oral Health. ;21(1):350. doi: 10.1186/s12903-021-01715-9. PMID: 34271907; PMCID: PMC8285804.

3M Transbond XT. 3M Transbond XT (Solventum) | Dental Product | Pearson Dental. (n.d.).

3M. Safety Data Sheet. (2020, July). https://multimedia.3m.com/mws/media/webserver?SSSSSuUn_zu8l00x4YtB5YtUPv70k17zHvu9lxtD7SSSSSS--

Conventional Resins used for Orthodontic Bonding

Mechanism:

- Etch to roughen the enamel surface
- Primer as a wetting agent
- Paste to adhere the bracket to the enamel surface mechanically

Clinical Implications:

- Transbond XT has **very high shear bonding strength** (12.91, $P < 0.001$) and performed best in vitro study by Tallani



RMGI Used for Orthodontic Bonding

Examples:

- RMGI: Fuji Ortho LC

Composition:

- Fluoroalumino-silicate glass (Powder), composite resin 2-hydroxyethyl methacrylate (HEMA), polyacrylic acid, and camphorquinone



RMGI Used for Orthodontic Bonding

Mechanism:

- Acid-base reaction between the liquid and powder (chelation)
- Light irradiation activates free radical polymerization of HEMA + two monomers -- > polyHEMA matrix that hardens the material
- After polymerization, resin monomers continue to self-cure

Clinical Implications:

- Potentially improved white spot lesion prevention in high risk individuals due to **fluoride release**
- **However**, some studies have found conflicting evidence and Masarykova 2023 noted that Fuji Ortho LC had **higher levels of microleakage** relative to 3 conventional resin bonding agents

Compomer used for Orthodontic Bonding

Examples:

- Dyract Cement

Composition: (In-general)

- Polyacid modified resin composite
- BisGMA and TEGDMA with fillers like silicate glass coated in silane.
- TCB 2-hydroxyethyl methacrylate with butane tetracarboxylic acid



Compomer used for Orthodontic Bonding

Mechanism:

- Light cured polymerization reaction followed by acid-base reaction

Clinical Implications:

- Fluoride release, but at lower levels than RMGI
- In vitro study by Jayasree 2019 concluded that Dyract had good bond strength and low microleakage

Giomer used for Orthodontic Bonding

Example: BeautiCem SA, a dual-cure resin cement with Glass ionomer filler

Composition:

- Resin matrix + surface pre-reacted glass ionomer (S-PRG) fillers



Giomer used for Orthodontic Bonding

Mechanism:

- Micromechanical bond to enamel (requires etch + bonding agent)
- Fluoride release and recharge via ion exchange from S-PRG fillers
- Provides composite-like strength with added bioactivity

Clinical Implications:

- Fluoride release may help prevent white spot lesions during orthodontic treatment
- Rusnac et al. (2019) reported sustained fluoride release and recharge from giomer materials, supporting their preventive effects against enamel demineralization during orthodontic treatment
- Lower bond strength than RMGI

BO 3:

Describe additional preventative strategies used in orthodontics (e.g., fluoride varnishes, sealants, glass ionomer cements, patient hygiene protocols, bonding area) and analyze how they complement adhesive selection.

Behavioral Objective 3

Search Strategy

PICO

P- Patients undergoing fixed orthodontic treatment

I- Use of preventive strategies

C- Orthodontic bonding without additional preventive measures

O- Reduced incidence of white spot lesions

Keywords

“Fluoride varnish” OR “fluoride application”

“Glass ionomer cement” OR “resin-modified glass ionomer”

“Sealant” OR “resin sealant”

“White spot lesions” OR “enamel demineralization”

Database Used

PubMed

Highest Level of Evidence

Systematic Review

BO 3: Describe additional preventive strategies used in orthodontics (e.g., fluoride varnishes, sealants, glass ionomer cements, patient hygiene protocols, bonding area) and analyze how they complement adhesive selection.

Overview

Preventive Strategies:

- ★ Fluoride Treatment (varnish, difluorosilane varnish, etc.)
- ★ Xylitol
- ★ Proper Oral Hygiene
- ★ Glass Ionomer Cements
- ★ Resin Modified Glass Ionomer Cement
- ★ Flash Free Brackets

How is demineralization measured in the referenced studies?

Are there White Spot Lesions present?
(yes or no)



What is the DIAGNOdent?
Measurement of demineralization
(0-12 = normal; >14 = carious)



Fluoride

Key: DFS

- difluorosilane

NaF

- sodium fluoride varnish

NaF+MI

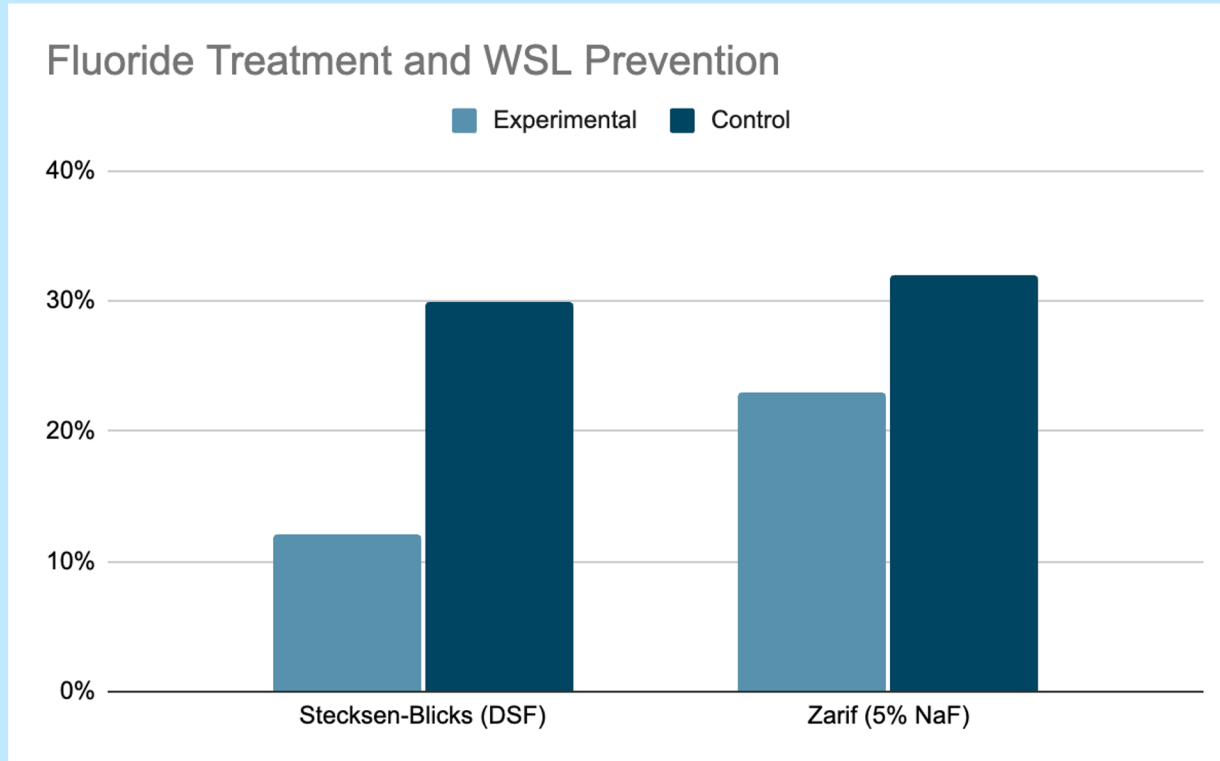
- sodium fluoride varnish +
10% casein phosphopeptide-
amorphous calcium
phosphate

AmF

- ammonium fluoride

First author, year/country	Age	Test/control	Duration	Dropout (%)	WSL score	Test [#]	Control	Frequency/applic. (n)	WSL prevalence T%/C%	WSL incidence T%/C%	P
Stecksen-Blicks, 2007/Sweden	12–15 years	137/136	Mean 1.7 years	5.00	Gorelick/photo	0.1% DFS	Placebo	6 weeks (n = 10)	12/30	7/26	<0.05
Rechmann, 2018/USA	13–26 years	20/20	12 months	7.50	EDI/clinical	5% NaF+MI ^A	Standard care ^B	3 months (n = 4)	54/55 ^C	32/41 ^C	NS
Sonesson, 2020/Sweden	12–18 years	85/81	Mean 1.7 years	10.80	Gorelick/photo	1.5% AmF	Placebo	6 weeks (n = 13)	42/44	12/26 ^D	NS/ <0.05
Zarif Najafi, 2021/Iran	12–18 years	30/30	12 months	3	Gorelick/clinical	5% NaF	Placebo	3 months (n = 4)	23/32 ^C	NR	<0.05
Flynn, 2022/USA	12–17 years	20/20	12 months	0	EDI/photo	5% NaF+MI	Sealants	4–6 weeks (n = 8– 12)	35/50	35/50	NS
Grocholewicz, 2022/Poland	16–50 years	30/30	12 months	0	Gorelick/clinical	5% NaF	Standard care	4 weeks (n = 12)	17/27	17/27	NS
Sardana, 2022/Hong Kong	13–25 years	33/33	18 months	10.60	Gorelick/photo	5% NaF	Standard care	4 weeks (n = 18)	10/18	NR	NS

Fluoride (cont.)



Xylitol

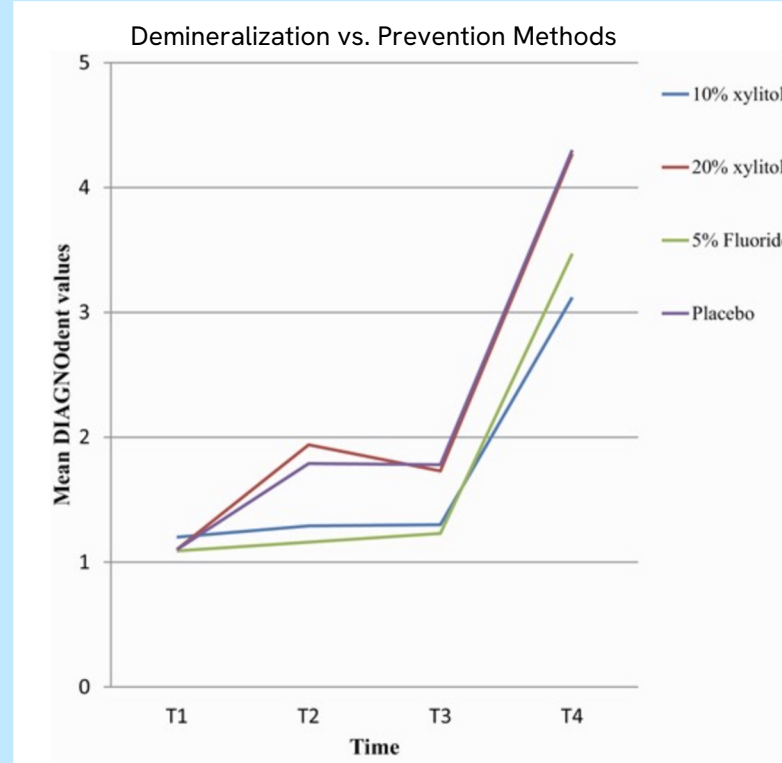
Key:

T1- before orthodontic treatment

T2- 3 months

T3- 6 months

T4- after orthodontic treatment



Is **xylitol** the future of white spot lesion prevention?

Oral Hygiene



Oral hygiene was strongly encouraged with patients undergoing orthodontic treatment. The regimen was brushing at least 2x per day with 1450 ppm fluoride toothpaste. Some studies also encouraged 0.05% sodium fluoride mouth rinse. There was no information on the addition of floss-threaders and/or a waterpik to get in between brackets in orthodontic devices.

Glass Ionomer



Glass ionomer cements actively release fluoride when they are set. With favorable performance in the presence of moisture, could be more effective in cases of difficult isolation (non-compliant patient, molar bands)

Resin Modified Glass Ionomer



Resin Modified Glass ionomer cements also actively release fluoride when they are set, but have an added resin filler to increase clinical ease with the ability to be light cured.

Mahmoud GA, Gordon PH, Pretty IA, McCabe JF, Hajeer MY. Effect of Fluoride Release on Enamel Demineralization Adjacent to Orthodontic Brackets. Cureus. 2023 Sep 28;15(9):e46132. doi: 10.7759/cureus.46132. PMID: 37779682; PMCID: PMC10538356.

El Helou M, Chakar S, Nicolas E, Estephan E, Cuisinier F, Barthélemy S. Can Orthodontic Adhesive Systems Inhibit the Formation and Development of White Spot Lesions During Fixed Orthodontic Treatment? A Systematic Review. J Adhes Dent. 2024 Oct 14;26:241-252. doi: 10.3290/j.jad.b5781299. PMID: 39397759; PMCID: PMC11748040.

Precoated Brackets



Precoated Flash Free Brackets have shown comparable plaque deposits, but different bacterial colonies that make-up the plaque.

What are they?

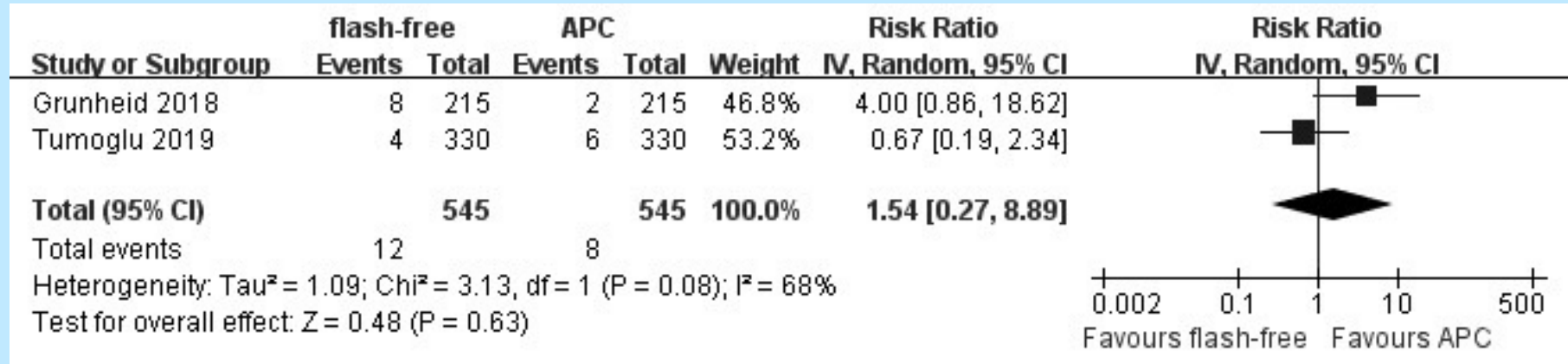
Coating: nonwoven, polypropylene fiber soaked in resin with low filler content

Fiber network is **compressible**, once force applied to bonding surface, resin fills gaps with minimal excess beyond bracket (no "flash")

Margins are much smoother compared to alternative bonding adhesives, which *may* contribute to the different plaque composition.

Future: incorporate fluoride-releasing adhesives (not currently offered)

Precoated Brackets and Bond Strength



NO statistical difference between conventional bonding strengths and precoated brackets

Systematic review comparing conventional bonding in precoated brackets vs. flash-free precoated brackets.

Demineralization of Enamel with Flash-Free Brackets (F) vs. Transbond XT Light Cure Adhesive (C)

(a commonly used conventional resin)

Demineralization measured using DIAGNOdent

Tooth-Side	Conventional (C)						Flash-Free (F)						C&F Comparison		
	T0		T1		T2		T0		T1		T2				
	Mean $\pm$ SD		Mean $\pm$ SD		Mean $\pm$ SD	T0-T1	T1-T2	T0-T2	Mean $\pm$ SD		Mean $\pm$ SD		T0-T1	T1-T2	T0-T2
U1-1	4.07 $\pm$ 2.03		4.87 $\pm$ 2.54		4.80 $\pm$ 2.63	ns	ns	ns	4.57 $\pm$ 2.03		4.73 $\pm$ 2.12		4.57 $\pm$ 1.68	ns	ns
U1-2	4.50 $\pm$ 2.43		3.50 $\pm$ 1.53		3.57 $\pm$ 2.73	ns	ns	0.02*	4.23 $\pm$ 1.68		3.63 $\pm$ 1.88		3.70 $\pm$ 1.93	ns	ns
U1-3	3.93 $\pm$ 2.29		3.80 $\pm$ 1.67		3.93 $\pm$ 1.78	ns	ns	ns	5.00 $\pm$ 2.0		4.20 $\pm$ 1.73		4.23 $\pm$ 1.98	ns	ns
U1-4	4.10 $\pm$ 2.16		4.23 $\pm$ 2.43		2.63 $\pm$ 0.93	ns	0.00*	0.00*	4.20 $\pm$ 1.37		4.03 $\pm$ 2.34		3.13 $\pm$ 2.37	ns	0.00*
U2-1	5.70 $\pm$ 3.09		4.47 $\pm$ 1.87		5.50 $\pm$ 2.70	ns	ns	ns	4.67 $\pm$ 1.79		5.10 $\pm$ 2.43		5.33 $\pm$ 2.59	ns	ns
U2-2	4.17 $\pm$ 1.98		3.60 $\pm$ 1.50		3.70 $\pm$ 2.09	ns	ns	ns	4.57 $\pm$ 1.94		3.27 $\pm$ 1.41		3.47 $\pm$ 1.72	0.01*	ns
U2-3	5.00 $\pm$ 2.74		4.60 $\pm$ 2.43		4.53 $\pm$ 2.01	ns	ns	ns	4.80 $\pm$ 1.58		4.63 $\pm$ 1.59		4.90 $\pm$ 1.81	ns	ns
U2-4	4.53 $\pm$ 2.19		4.50 $\pm$ 2.53		3.33 $\pm$ 1.75	ns	ns	0.01*	4.43 $\pm$ 1.43		4.43 $\pm$ 2.3		3.77 $\pm$ 2.51	ns	0.02*
U3-1	5.67 $\pm$ 2.96		5.83 $\pm$ 2.93		6.73 $\pm$ 4.65	ns	ns	ns	5.60 $\pm$ 2.75		5.03 $\pm$ 2.24		5.63 $\pm$ 2.46	ns	ns
U3-2	5.43 $\pm$ 2.70		4.60 $\pm$ 2.24		5.27 $\pm$ 2.97	ns	ns	ns	5.40 $\pm$ 2.31		4.20 $\pm$ 1.32		4.47 $\pm$ 2.00	0.02*	ns
U3-3	5.17 $\pm$ 2.60		5.40 $\pm$ 2.11		5.93 $\pm$ 2.68	ns	ns	ns	4.80 $\pm$ 1.75		5.30 $\pm$ 1.95		5.73 $\pm$ 2.86	ns	ns
U3-4	4.97 $\pm$ 2.76		4.77 $\pm$ 2.53		3.87 $\pm$ 1.28	ns	ns	ns	5.07 $\pm$ 2.42		4.67 $\pm$ 2.4		3.57 $\pm$ 1.25	ns	0.01*
U4-1	7.97 $\pm$ 3.13		5.93 $\pm$ 2.64		7.23 $\pm$ 2.78	0.02*	ns	ns	7.07 $\pm$ 2.94		5.03 $\pm$ 2.11		6.10 $\pm$ 3.09	0.02*	ns
U4-2	5.90 $\pm$ 3.07		5.07 $\pm$ 2.21		6.07 $\pm$ 3.25	ns	ns	ns	6.07 $\pm$ 3.02		4.40 $\pm$ 1.25		6.20 $\pm$ 4.87	0.01*	ns
U4-3	5.77 $\pm$ 2.34		5.07 $\pm$ 2.32		5.80 $\pm$ 2.66	ns	ns	ns	5.87 $\pm$ 2.73		5.17 $\pm$ 2.49		6.80 $\pm$ 3.81	ns	ns
U4-4	5.77 $\pm$ 2.78		4.63 $\pm$ 2.74		3.70 $\pm$ 2.77	ns	ns	0.00*	5.23 $\pm$ 2.24		4.60 $\pm$ 2.58		4.40 $\pm$ 4.39	ns	0.01*
U5-1	8.13 $\pm$ 3.36		6.40 $\pm$ 2.72		8.13 $\pm$ 3.08	0.04*	0.03*	ns	7.80 $\pm$ 2.99		5.97 $\pm$ 2.34		7.83 $\pm$ 2.87	ns	0.01*
U5-2	7.30 $\pm$ 3.19		4.90 $\pm$ 2.12		6.63 $\pm$ 3.23	0.03*	0.04*	ns	7.17 $\pm$ 2.65		5.20 $\pm$ 2.63		6.63 $\pm$ 3.19	0.00*	ns
U5-3	5.93 $\pm$ 2.41		4.70 $\pm$ 1.37		6.67 $\pm$ 3.37	0.04*	ns	ns	6.40 $\pm$ 2.33		4.77 $\pm$ 2.14		6.13 $\pm$ 2.87	0.00*	ns
U5-4	5.70 $\pm$ 2.53		3.93 $\pm$ 1.57		3.87 $\pm$ 1.55	0.01*	ns	0.001*	5.57 $\pm$ 2.42		4.43 $\pm$ 2.19		4.10 $\pm$ 2.78	ns	0.00*
L1-1	5.37 $\pm$ 2.19		4.67 $\pm$ 2.12		6.33 $\pm$ 3.10	ns	0.02*	ns	5.77 $\pm$ 2.39		4.70 $\pm$ 1.70		6.33 $\pm$ 2.92	ns	0.02*
L1-2	5.13 $\pm$ 2.78		4.43 $\pm$ 2.13		5.27 $\pm$ 2.79	ns	ns	ns	3.90 $\pm$ 1.63		4.40 $\pm$ 2.43		4.93 $\pm$ 2.61	ns	ns
L1-3	6.00 $\pm$ 1.97		4.87 $\pm$ 2.16		6.37 $\pm$ 2.47	0.01*	0.00*	ns	5.90 $\pm$ 1.97		4.60 $\pm$ 1.81		5.93 $\pm$ 2.50	0.002*	0.014*
L1-4	4.07 $\pm$ 1.26		2.90 $\pm$ 1.49		3.23 $\pm$ 1.65	ns	0.00*	0.00*	3.90 $\pm$ 1.83		3.23 $\pm$ 1.77		3.87 $\pm$ 2.98	0.02	ns
L2-1	5.47 $\pm$ 2.47		4.73 $\pm$ 2.48		6.30 $\pm$ 2.68	ns	0.02*	ns	5.57 $\pm$ 2.54		4.97 $\pm$ 2.19		6.37 $\pm$ 2.88	ns	ns
L2-2	4.33 $\pm$ 1.71		4.73 $\pm$ 2.18		5.23 $\pm$ 2.34	ns	ns	ns	4.97 $\pm$ 2.14		4.37 $\pm$ 1.54		5.33 $\pm$ 3.13	ns	ns
L2-3	5.70 $\pm$ 2.38		5.17 $\pm$ 2.05		7.73 $\pm$ 7.24	ns	ns	ns	5.70 $\pm$ 2.07		5.00 $\pm$ 1.60		6.57 $\pm$ 3.37	ns	ns
L2-4	4.03 $\pm$ 1.19		3.20 $\pm$ 1.85		3.80 $\pm$ 1.71	0.02*	ns	ns	3.90 $\pm$ 1.54		3.53 $\pm$ 2.00		3.80 $\pm$ 2.37	0.03*	ns
L3-1	4.57 $\pm$ 1.98		4.77 $\pm$ 2.25		5.83 $\pm$ 2.91	ns	0.04*	ns	5.13 $\pm$ 2.36		5.67 $\pm$ 2.47		5.93 $\pm$ 2.79	ns	ns
L3-2	4.93 $\pm$ 2.36		4.00 $\pm$ 1.62		6.30 $\pm$ 6.68	ns	0.03*	ns	4.27 $\pm$ 1.98		4.40 $\pm$ 2.34		4.80 $\pm$ 2.19	ns	ns
L3-3	5.07 $\pm$ 1.96		4.57 $\pm$ 1.89		5.90 $\pm$ 3.10	ns	ns	ns	5.20 $\pm$ 2.25		5.23 $\pm$ 2.06		5.93 $\pm$ 2.95	ns	ns
L3-4	4.13 $\pm$ 1.33		3.40 $\pm$ 0.81		3.97 $\pm$ 2.27	0.03*	ns	ns	4.13 $\pm$ 1.43		3.73 $\pm$ 2.48		3.30 $\pm$ 1.09	0.02*	0.02*
L4-1	4.97 $\pm$ 2.06		4.53 $\pm$ 1.43		5.30 $\pm$ 1.86	ns	ns	ns	5.23 $\pm$ 1.83		5.13 $\pm$ 2.01		6.03 $\pm$ 3.01	ns	ns
L4-2	4.97 $\pm$ 2.41		4.83 $\pm$ 2.17		6.13 $\pm$ 2.60	ns	0.02*	ns	5.17 $\pm$ 2.35		5.37 $\pm$ 2.09		5.93 $\pm$ 2.32	ns	ns
L4-3	4.90 $\pm$ 2.23		4.43 $\pm$ 1.38		6.20 $\pm$ 4.66	ns	0.03*	ns	4.80 $\pm$ 1.75		4.63 $\pm$ 1.79		6.07 $\pm$ 4.26	ns	ns
L4-4	4.27 $\pm$ 1.89		2.87 $\pm$ 1.36		3.20 $\pm$ 1.30	0.00*	ns	0.02*	4.10 $\pm$ 1.73		3.07 $\pm$ 1.72		3.03 $\pm$ 1.38	0.001*	0.001*
L5-1	5.27 $\pm$ 2.65		4.90 $\pm$ 1.71		5.67 $\pm$ 4.60	ns	ns	ns	5.63 $\pm$ 2.51		5.03 $\pm$ 2.13		6.13 $\pm$ 3.49	ns	ns
L5-2	6.30 $\pm$ 2.85		5.07 $\pm$ 2.33		7.03 $\pm$ 2.99	ns	0.00*	ns	6.77 $\pm$ 2.43		5.23 $\pm$ 2.56		7.27 $\pm$ 3.83	0.02*	0.03*
L5-3	5.60 $\pm$ 3.09		4.30 $\pm$ 1.29		5.63 $\pm$ 2.91	ns	ns	ns	5.10 $\pm$ 2.06		4.47 $\pm$ 1.85		5.70 $\pm$ 3.64	ns	ns
L5-4	4.47 $\pm$ 2.45		3.03 $\pm$ 0.96		3.13 $\pm$ 1.46	0.01*	ns	0.01*	3.83 $\pm$ 1.56		3.10 $\pm$ 1.79		3.30 $\pm$ 1.70	0.04*	ns

U: Upper; L: Lower.

First number indicates tooth number.

Second number indicates bracket sides 1: Distal; 2: Gingival; 3: Mesial; 4: Incisal/Occlusal.

SD: Standard deviation; * demonstrates statistically significant increase; ns: non-significant.

* $P < .05$.

BO 4:

Compare and contrast the advantages and disadvantages of fluoride-releasing adhesives and adjunctive preventative methods using current scientific evidence.

Behavioral Objective 4

Search Strategy

PICO

P- Patients undergoing fixed orthodontic treatment

I- Fluoride-releasing adhesives

C- Adjunctive methods

O- Development of white spot lesions

Keywords

"fluoride-releasing adhesive"

"glass ionomer cement"

"RMGI" OR "compomer" OR "nanoparticle orthodontic adhesive"

"fluoride varnish" OR "sealant" OR "oral hygiene protocol"

Database Used

PubMed

Highest Level of Evidence

Systematic Review

Fluoride releasing adhesives: *the good, the bad, and the ugly*

- ★ Fluoride-releasing adhesives:
 - provide fluoride delivery to the bracket-enamel interface during fixed orthodontic treatment
- ★ Common examples:
 - conventional glass ionomer cement (GICs)
 - resin-modified glass ionomer cement (RMGICs)
 - polyacid-modified composite resins (PMCRs)

Fluoride releasing adhesives: *research*

★ Overview of study 1

- In-vitro experimental study (2023)
- Goal: comparing fluoride-releasing vs. non-fluoride orthodontic adhesives
- Materials tested – *Fuji Ortho LC, Dyract cement, Ketac cement, Transbond*
- Two groups of extracted human premolars, free from any decay/defects
 - Group one - pH cycle (mimics remin/demin cycle)
 - Group two - pH cycle + *fluoride exposure*



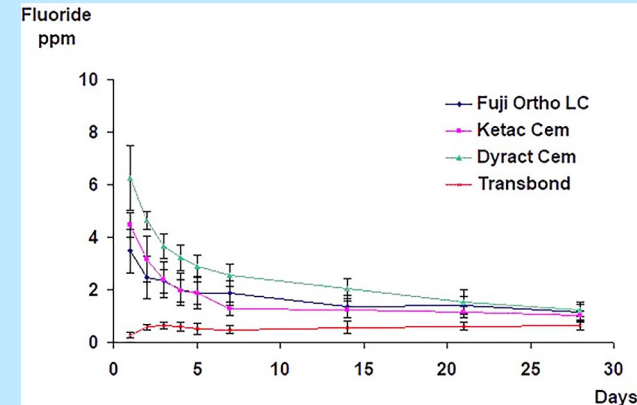
★ Overview of study 2

- Systematic review (2023)
 - Only *in-vivo* and clinical trials included
 - 12 studies total
- Goal: assess whether orthodontic bonding systems prevent orthodontic-induced white spot lesions



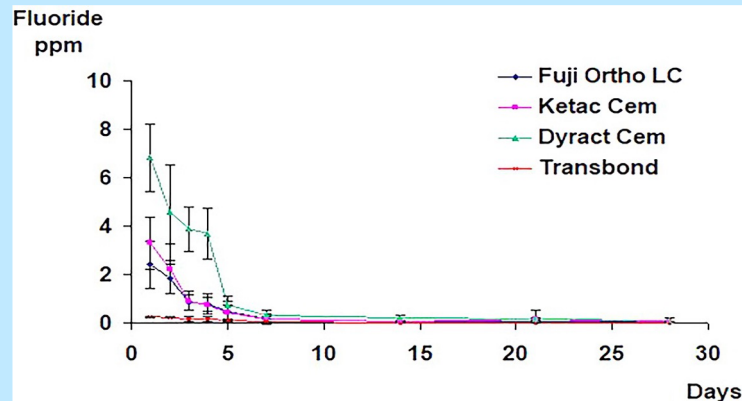
Fluoride releasing adhesives: *advantages*

- ★ When fluoride releasing adhesives are used in tandem with other fluoride products (rinse, gels, toothpastes), fluoride release is enhanced.
 - Adhesives exposed to fluoride had a higher fluoride release curve and higher constant levels 1 week later.
 - Dyract Cement (compomer) > Ketac Cement (RMGI) = Fuji Ortho LC (RMGI)
 - Mechanism: sustained fluoride diffusion → enhanced enamel resistance → inhibition of crystal dissolution
- ★ Preventing enamel demineralization and protecting against acid attack
 - Control = Transbond (composite resin)
 - Effects on fluoride release were negligible
 - passive chemical defense
 - best combined with topical fluoride varnish or high-fluoride toothpaste to maintain long-term efficacy



Fluoride releasing adhesives: *disadvantages*

- ★ Fluoride burst - first 24 hours
 - Preventative until significant drop off 1 week later to a constant level
- ★ Retention
 - Balance between reducing WSLs and bracket retention
 - Although addition of antimicrobials is promising, they reduce bond performance

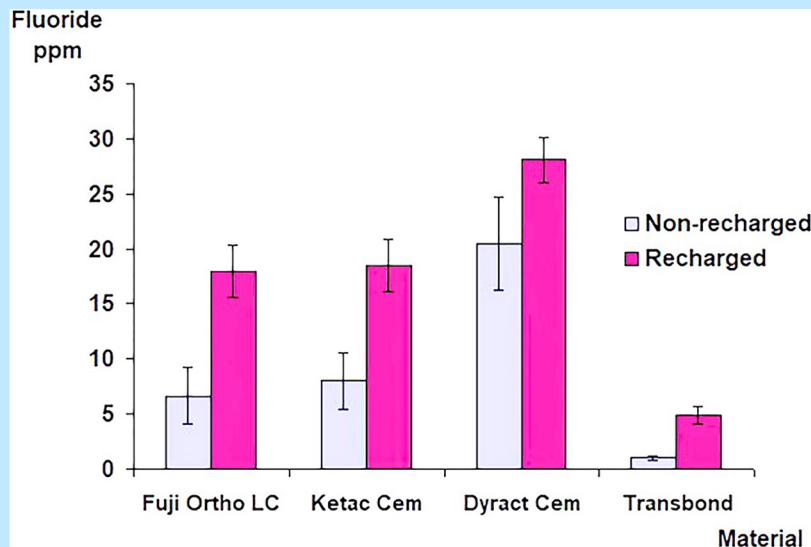


Mahmoud GA, Gordon PH, Pretty IA, McCabe JF, Hajeer MY. Effect of Fluoride Release on Enamel Demineralization Adjacent to Orthodontic Brackets. Cureus. 2023 Sep 28;15(9):e46132. doi: 10.7759/cureus.46132. PMID: 37779682; PMCID: PMC10538356.

El Helou M, Chakar S, Nicolas E, Estephan E, Cuisinier F, Barthélemy S. Can Orthodontic Adhesive Systems Inhibit the Formation and Development of White Spot Lesions During Fixed Orthodontic Treatment? A Systematic Review. J Adhes Dent. 2024 Oct 14;26:241-252. doi: 10.3290/j.jad.b5781299. PMID: 39397759; PMCID: PMC11748040.

Is there one that out-competes?

1. **Dyract Cement (compomer)** 2. Ketac Cement, Fuji Ortho LC (RMGI) 3. Transbond (composite)



Adjunctive therapies and success?

★ Calcium Fluoride nanoparticle primer

- Advantages: reduces demineralization, suppresses S. mutans at bonding
- Disadvantages: no significant WSL prevention at debonding, cytotoxicity, and bond strength
- Experimentally created primer... *not commercially available*
 - Nanoparticles + Transbond XT



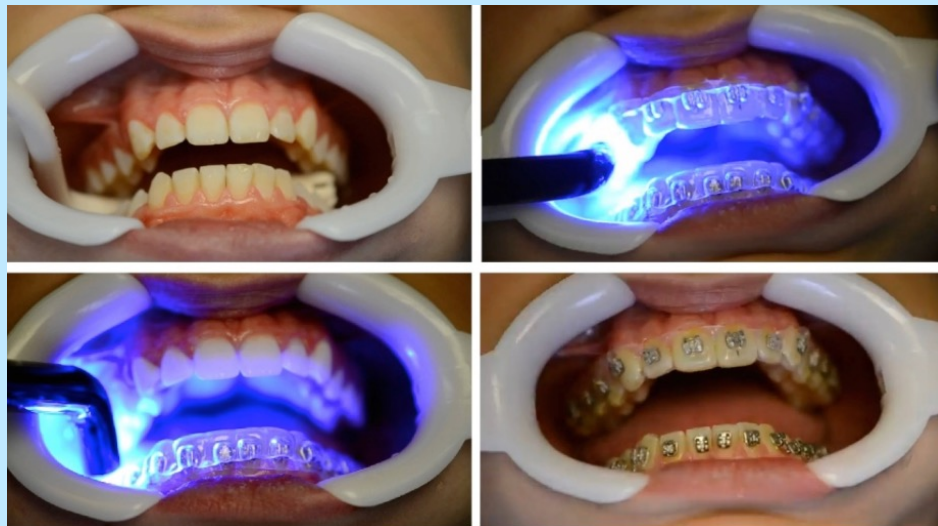
★ Amorphous Calcium Phosphate-containing adhesives

- Effective in minimizing enamel demineralization
 - At acidic pH, ACP converts to hydroxyapatite
- Concern with affecting bone adhesion
- Aegis - light-cured adhesive
- Recaldent (CPP-ACP) - MI paste
- Enamelon - gel



Adjunctive therapies and success?

- ★ Indirect bonding with flowable composites
 - More effective in preventing WSLs than direct composite bonding
 - Reason due to precise bracket placement not fluoride release



Adjunctive therapies and success?

<https://www.gc.dental/ortho/en-US/products/gcorthoconnect>

- ★ Self- adhesive ortho resin (GC Ortho Connect)
 - Conflicting studies:
 - Pro - decreased residual adhesive on enamel after debonding averts WSL formation
 - Against - no primer leads to increased WSL formation

One-step system

No need to apply primer or adhesive on the tooth, as it is already incorporated into the paste.

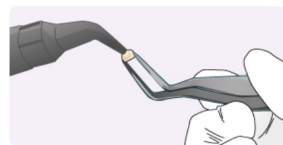
A simple etching of the enamel followed by drying is enough to prepare the bracket or buccal tube for bonding.



1. Clean tooth surfaces with pumice and water. Rinse thoroughly with water and dry.



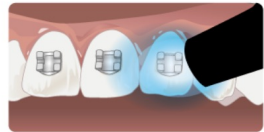
2. Etch the tooth surfaces with GC Ortho Etching Gel for 30 sec., rinse and dry. Porcelain, metal-, composite-restored teeth should be treated with the appropriate conditioner.



3. Coat the bonding surface of the bracket with GC Ortho Connect paste.



4. Remove the excess adhesive using an explorer or scaler.



5. Cure each bracket for 10 seconds mesially and distally with an LED curing light.

Excess can be easily removed under near UV-light for a more accurate cleaning.

Adjunctive therapies and success?

- ★ TiO₂ nanoparticles
 - Advantages: antimicrobial properties that decrease *S. mutans*
 - Disadvantages: nanoparticles can cause oxidative stress, histopathological changes, and carcinogenicity
 - Experimental additive... *not commercially available*
- ★ Fluoride releasing primers (Opal Seal, Clearfil Protect Bond)
 - Lacked ability to prevent WSLs
 - More studies needed in optimizing fluoride concentration and release kinetics
 - Opal Seal Manufacturer Claims
 - Releases and recharges fluoride
 - Nonyellowing and stain resistant
 - Drying agent that seeks out moisture and draws resin in



Adjunctive therapies and success?

3M™ Vanish™ 5% Sodium Fluoride White Varnish

- Delivers targeted- and extended-release fluoride and calcium
- Easy and fast to apply
- No waiting to eat or drink
- Can be applied to moist teeth
- Adheres to the teeth until brushed off



3M™ Vanish™ XT Extended Contact Varnish

- Unique resin-modified glass ionomer material that releases fluoride, calcium and phosphate
- Creates a protective barrier against demineralization and acid erosion
- Protects for up to 6 months



5,000 ppm sodium fluoride toothpaste

3M™ Clinpro™ 5000 1.1% Sodium Fluoride Anti-Cavity Toothpaste

- Delivers 5,000 ppm of fluoride, calcium and phosphate (FTCP)
- Proven to help reverse white spot lesions
- Low abrasion for exposed root surfaces
- Excellent patient compliance—no change in brushing routine



Vanilla Mint and Spearmint flavors

For gingivitis: antimicrobial rinses

3M™ PerioMed™ 0.63% Stannous Fluoride Oral Rinse

- For decay prevention, sensitivity and gingival inflammation
- Supports tooth and gingival health
- Antimicrobial: effective against bacteria for up to 8 hours



Clinical Reality...

Prevention is *highly dependent* on patient compliance.

- ★ The best approach during orthodontic treatment is to prevent lesions from forming. It has been concluded that fluoride delivery, oral hygiene compliance, and dietary control significantly reduce demineralization.

Future research?

- ★ Level of fluoride release needed to prevent enamel demineralization is still unknown.
- ★ Nanoparticles incorporated into adhesives or primers exhibit promise in preventing white spot lesions.
 - In need of a standardized way to measure effectiveness and safety
 - The effectiveness in preventing WSLs should balance biocompatibility, bond strength and demineralization control.

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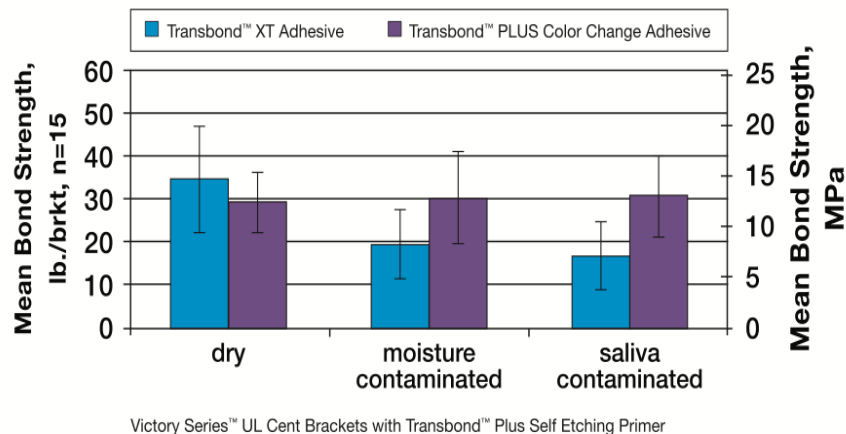
Future research?

★ Bond strength

- Experimental study (2022) comparing extracted human premolars
- Bonding materials tested:
 - Fluoride-releasing adhesive= Transbond™ Plus Color Change Adhesive
 - Conventional adhesive = Transbond™ XT
- Main finding:
 - Fluoride releasing adhesives → **decrease** in shear bond strength, especially in eroded enamel
- Conclusion:
 - Clinical decision guided by importance... caries prevention or bond strength?



Comparative Bond Strength



What are we trying to avoid?

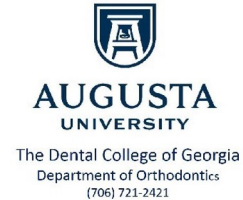
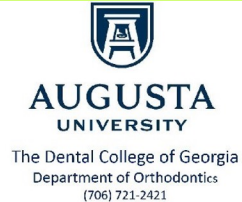


Case Introduction

Pt. started comprehensive treatment at age 14 and ended at age 18.

Med history includes autism and bad oral hygiene.

Case Control Study- no fluoride in cement



Case- Before Treatment



AUGUSTA
UNIVERSITY

The Dental College of Georgia
Department of Orthodontics
(706) 721-2421



Case- Before Treatment



Case- After



Case- After



3M Science.
Applied to Life.™

3M™ Clinpro™ XT Varnish
Durable Fluoride-Releasing Coating



Creation of a protective barrier around orthodontic brackets

Jacqueline Esch, Munich, Germany

3M Fluoride Varnish Procedure



Figure 1: Patient with orthodontic brackets shortly after their placement. The tooth surface around the brackets is cleaned with a rubber brush.



Figure 2: For optimal removal of plaque and debris, 3M™ Clinpro™ Prophyl Paste may be used. A cleaning medium containing oils is not indicated.



Figure 3: Thorough removal of the polishing paste by rinsing with water.



Figure 4: Removal of pooled water on the surfaces around the orthodontic brackets with air. The tooth should still be slightly moist afterwards.

3M Fluoride Varnish Procedure (cont.)



Figure 5: Application of 35% phosphoric acid etchant to the enamel surfaces. The recommended etching time is 15 to 60 seconds.



Figure 6: Thorough rinsing with air/water spray for complete removal of the etchant.



Figure 7: A small amount of 3M™ Clinpro™ XT Varnish Durable Fluoride-Releasing Coating is dispensed onto the mixing pad. The 3M™ Clicker™ Dispenser ensures the correct mixing ratio is obtained.

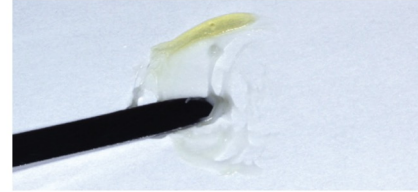


Figure 8: Mixing paste and liquid components with a small spatula for a homogeneous result with smooth consistency. The optimal mixing time is 10 to 15 seconds



Figure 9: Application of the glossy coating to the vestibular tooth surfaces around the orthodontic brackets with a brush. The layer should not be thicker than 0.5 mm. Contact with the soft tissues should be avoided.



Figure 10: Light curing of each area for 20 seconds. In areas without exposure to the curing light, the delayed auto-setting mechanism of the coating ensures a complete cure.

3M Fluoride Varnish Outcome

Outcome

The result of this treatment is a protective coating of the tooth surfaces with increased risk for caries or white spot lesions. For an ideal effect, the application should be repeated after six months. At any time, it is possible to remove the coating by cleaning the tooth surfaces with a coarse prophylaxis paste.

In our experience, the described application of 3M™ Clinpro™ XT Varnish Durable Fluoride-Releasing Coating is an effective measure as a site-specific protective coating for high-risk, caries-prone areas, including around fixed orthodontic brackets and acid erosion-prone surfaces.

www.3M.com



BO 5:

Apply a clinical decision tree to select appropriate preventative measures for white spot lesion management based on patient risk factors & treatment context

Behavioral Objective 5

Risk Assessment



Low Risk

Age (old enough to perform OH regularly)
Diet (low carbohydrates, lots of water, etc.)
Low CRA and current carious activity
Access to proper oral hygiene instruments



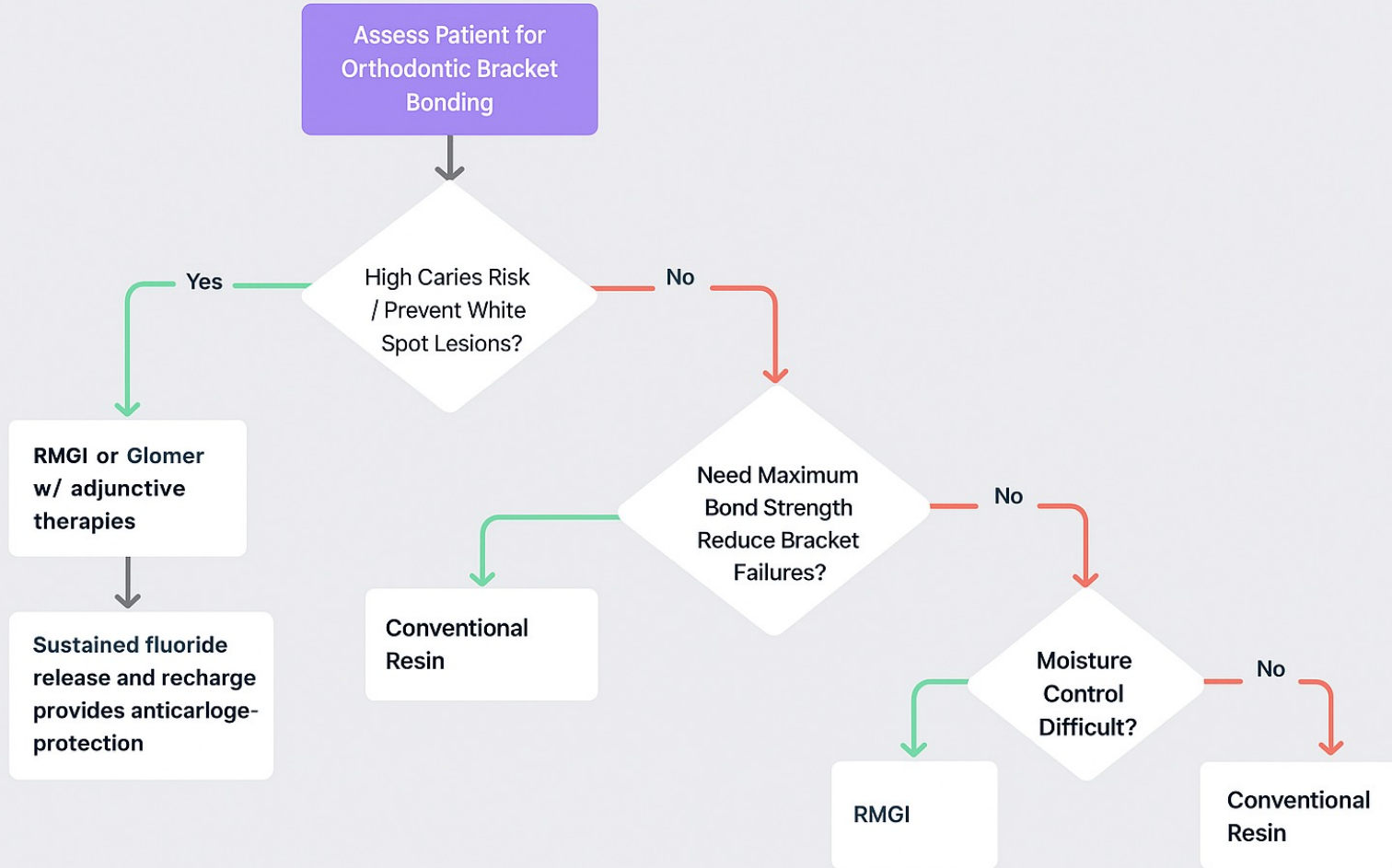
Moderate Risk

Age (younger, but performs OH mostly regularly)
Diet (moderate carbohydrates, mostly of water, etc.)
Moderate CRA and mild current carious activity
Access to proper oral hygiene instruments



High Risk

Age (young, cannot perform OH regularly)
Diet (high carbohydrates, no water, etc.)
High CRA and extensive carious activity
No access to proper oral hygiene instruments





6 Thinking Hats



**Green Hat - Creativity**

Focuses on new ideas, alternatives, and innovation. Think outside the box and suggest fresh possibilities

**White Hat - Facts & Information**

Looks at data, evidence, and facts only. What do we know? What information do we still need?

**Red Hat - Feelings & Intuition**

Expresses emotions, gut feelings, and instincts. No need for logic, just honest reactions

**Black Hat - Caution & Risk**

Identifies potential problems, weaknesses, and reasons an idea might not work. Encourages critical thinking

**Yellow Hat - Optimism & Benefits**

Highlights positive outcomes, opportunities, and benefits. Looks at why an idea could work

**Blue Hat - Process & Control**

Manages the discussion. Keeps everyone focused, summarizes key points, and decides next steps





A 15-year-old orthodontic patient, Emily, has had braces for 6 months.

At her check-up, her orthodontist notices early signs of white spot lesions forming around several brackets.

The dental team gathers to discuss how to prevent further lesions and manage the early ones using the **Six Thinking Hats** approach.



In-Class Activity: Based on Heads Up!©

Each table will receive a handout with a summary of the case scenario and description of the Six Thinking Hats.

Face-down cards will be randomly assigned to 6 members of each table.

Work as a team to describe the thought processes and/or emotions associated with any stakeholder of the case (dental team, patient, parent, manufacturer) in the context of each hat. First team to decipher all 6 wins a prize!

**Green Hat - Creativity**

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White Hat: Facts and Information (*Faculty, Researcher, Orthodontist*)

- White spot lesions result from demineralization due to plaque accumulation around brackets.
- Risk factors include poor oral hygiene, high-sugar diet, and long treatment duration.
- Preventive strategies: fluoride varnish, fluoride toothpaste, glass ionomer bonding materials, patient education, dietary counseling, and regular professional cleanings.
- Emily's brushing habits are inconsistent, and she consumes sugary snacks frequently.



Red Hat (Feelings & Intuition *(Patient, Parent, Orthodontist)*

- The orthodontist feels concerned that Emily may lose motivation or confidence if white spots worsen.
- The hygienist feels frustrated because they've already emphasized oral hygiene before.
- Emily feels embarrassed and discouraged after being told she might have permanent marks.
- Parents feel guilty they haven't monitored her brushing closely.



Black Hat: Caution & Risks (*Orthodontist*)

- Overusing fluoride varnish could lead to uneven remineralization or a false sense of security.
- Switching to a new bonding material mid-treatment might risk bracket failure.
- If white spots progress to cavitation, restorative treatment will be required post-orthodontics.
- Patient compliance may remain low even with repeated counseling



Yellow Hat: Benefits & Value (*Sales Rep, Office Marketing Rep*)

- Implementing a structured preventive plan can stop progression and even reverse early lesions.
- Using RMGI or fluoride-releasing sealants can offer long-term protection.
- Reinforcing education could empower Emily to take control of her oral health.
- Early detection allows for intervention before permanent damage occurs.



Green Hat: Creativity & Alternatives (*Manufacturer, Sales Rep, Provider's Colleague*)

- Introduce a reward system or mobile app to track daily brushing.
- Balance the treatment goals and OHI difficulties by using clear aligners.
- Use disclosing tablets to show Emily where plaque accumulates.
- Enroll Emily in a case study testing Nanoparticles in adhesive for preventing white spot lesions.



Blue Hat: Process & Big Picture (*Provider, ADA, AJO-DO*)

The team agrees to a multi-step plan:

1. Educate Emily and parents again using visual aids.
2. Apply fluoride varnish at every visit.
3. Recommend fluoride toothpaste and mouthrinse.
4. Reinforce proper brushing technique with disclosing tablets.
5. Monitor progress at each appointment.

The blue hat leader will ensure consistent follow-up and evaluate results after 3 months.



BO Summary Slide

BO 1: Identify the biological and clinical factors that contribute to the formation of white spot lesions around orthodontic brackets.

Biological Factors

- Plaque rich in *Strep Mutans*, *Lactobacillus*
- Imbalance of remin/demin cycle
- Low Fluoride availability
- Systemic host factors

Clinical Factors

- Patient Compliance**
- Bracket Type, Bonding Procedures, Material Choice
- Fluoride exposure
- Duration of Treatment

BO 2: Differentiate between conventional bonding agents and fluoride-releasing adhesives by explaining their composition, mechanism of action, and their clinical implications.

- **Conventional** bonding agents use resin monomers, solvents, and a polymerization system to form a hybrid layer through etch, prime, and bond steps.
- **Fluoride-releasing** adhesives contain fluoride fillers or glass ionomer components that provide both micromechanical and chemical bonding while releasing fluoride to prevent caries. They're ideal for patients with high caries risk during orthodontic bonding.

BO Summary Slide

BO 3: Describe additional preventive strategies used in orthodontics (e.g., fluoride varnishes, sealants, glass ionomer cements, patient hygiene protocols, bonding area) and analyze how they complement adhesive selection.

Preventive measures like fluoride varnish, sealants, glass ionomer cements, proper hygiene, and clean bonding areas protect enamel during orthodontic treatment. These methods strengthen enamel, prevent white spot lesions, and enhance adhesive durability by reducing demineralization.

BO 4: Compare and contrast the advantages and disadvantages of fluoride-releasing adhesives and adjunctive preventive methods using current scientific evidence.

Fluoride-releasing adhesives, when used with other topical fluoride products, have been proven to be advantageous at preventing WSLs, with all other factors held constant. However, there is an **inverse relationship between reducing WSLs and bracket retention** that clinicians must consider. While some adjunct therapies have had success, ultimately the treatment of choice is dependent on patient compliance.

BO Summary Slide

BO 5: Apply a clinical decision tree to select appropriate preventive measures for white spot lesion management based on patient risk factors and treatment context.

Appropriate preventative measures begins with **assessing the patient's caries risk and implementing a stepwise approach.**

- Low-risk:** Maintain good hygiene with fluoride toothpaste and rinses plus regular cleanings.
- Moderate-risk:** Add fluoride varnish and sealants around brackets; reinforce brushing and diet control.
- High-risk:** Use fluoride-releasing adhesives or glass ionomer cements, frequent varnish or CPP-ACP, sealants, and prescription fluoride. Manage white spot lesions with remineralizing agents or resin infiltration as needed.

References



Thank you!
